

28/07/26

SIMATS Online Programming Lab
PYTHON PROGRAMMING

Suba Shree K
192521396RD

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Q1 Void Function (No Return Value) 5 marks

Write a Python program to define a function that displays "Welcome to Python Programming".

Your Code

```
1 def welcome():  
2     print("Welcome to Python Programming")  
3 welcome()
```

Input
Welcome to Python Programming

Output
Welcome to Python Programming

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Q2 Void Function (No Return Value) 5 marks

Write a function to print the multiplication table of a given number.

Your Code

```
1 def table(n):  
2     for i in range(1, 11):  
3         print(n, "x", i, "=", n * i)  
4 num = int(input("Enter a number: "))  
5 table(num)
```

Input
5

Output

```
Enter a number: 5 x 1 = 5  
5 x 2 = 10  
5 x 3 = 15  
5 x 4 = 20  
5 x 5 = 25  
5 x 6 = 30  
5 x 7 = 35  
5 x 8 = 40  
5 x 9 = 45  
5 x 10 = 50
```

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Q3 Void Function (No Return Value) 5 marks

Write a function to print all even numbers from 1 to N.

Your Code

```
1 def even_numbers(n):  
2     for i in range(1, n + 1):  
3         if i % 2 == 0:  
4             print(i)  
5 n = int(input("Enter N: "))  
6 even_numbers(n)
```

Input
10

Output

```
Enter N: 2  
4  
6  
8  
10
```

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Q4 Fruitful Function (Returns One Value)

5 marks

Write a function that accepts a string and returns the number of vowels in it.

Your Code

```
1 def count_vowels(s):
2     count = 0
3     for ch in s:
4         if ch.lower() in "aeiou":
5             count += 1
6     return count
7 text = input("Enter a string: ")
8 print("Number of vowels:", count_vowels(text))
```

Input

Python Programming

Output

Enter a string: Number of vowels: 4

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Q5 Fruitful Function (Returns One Value)

5 marks

Write a function that accepts a list of integers and returns the largest element.

Your Code

```
1 def largest(lst):
2     m = lst[0]
3     for i in lst:
4         if i > m:
5             m = i
6     return m
7 lst = list(map(int, input().split()))
8 print(largest(lst))
```

Input

26
16
7
10

Output

26

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Q6 Fruitful Function (Returns One Value)

5 marks

Write a function that accepts a number and returns whether it is prime or not.

Your Code

```
1 def is_prime(n):
2     if n < 2:
3         return False
4     for i in range(2, n):
5         if n % i == 0:
6             return False
7     return True
8 num = int(input("Enter a number: "))
9 if is_prime(num):
10     print("Prime")
11 else:
12     print("Not Prime")
```

Input

7

Output

Enter a number: Prime

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Q7 Function Returning Two or More Values

5 marks

Write a function that accepts a student's marks in three subjects and returns the total, average, and grade.

Your Code

```
1 def result(a, b, c):
2     total = a + b + c
3     avg = total / 3
4
5     if avg >= 90:
6         grade = "S"
7     elif avg >= 75:
8         grade = "A"
9     elif avg >= 50:
10        grade = "B"
11    else:
12        grade = "F"
13    return total, avg, grade
14 a = int(input())
15 b = int(input())
16 c = int(input())
17 t, av, g = result(a, b, c)
18 print(t)
19 print(av)
20 print(g)
```

Input

98
90
86

Output

274
91.33333333333333
S

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Q8 Function Returning Two or More Values

5 marks

Write a function that accepts the radius of a circle and returns its area and circumference.

Your Code

```
1 def circle(r):
2     area = 3.14 * r * r
3     circum = 2 * 3.14 * r
4     return area, circum
5 r = float(input())
6 a, c = circle(r)
7 print(a)
8 print(c)
```

Input

5

Output

78.5
31.400000000000002

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Q9 Function Returning Two or More Values

5 marks

Write a function that accepts a number and returns its square, cube, and square root.

Your Code

```
1 def values(n):
2     sq = n * n
3     cube = n * n * n
4     root = n ** 0.5
5     return sq, cube, root
6 n = int(input())
7 s, c, r = values(n)
8 print(s)
9 print(c)
10 print(r)
```

Input

6

Output

36
216
2.449489742783178

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Q10 Keyword Arguments

5 marks

Write a function student(name, age, department) and call it using keyword arguments.

Your Code

```
1 def student(name, age, department):
2     print("Name:", name)
3     print("Age:", age)
4     print("Department:", department)
5
6 student(name="mohan", age=19, department="B.Tech IT")
```

Input

stdin input

Output

Name: mohan
Age: 19
Department: B.Tech IT